

End-to-End Semantic Segmentation Pipeline for Deforestation Monitoring Using Sentinel-2 and Foundation Models

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Abstract—Deforestation monitoring is a critical challenge for environmental protection, biodiversity conservation, and climate change mitigation. Satellite-based Earth observation provides a scalable solution, but accurate and timely detection of forest loss requires robust semantic segmentation models and reliable annotated data. A major limitation in this domain is the cost and complexity of producing high-quality ground-truth labels at scale.

This project presents Deforested Patch Evolution, an end-to-end semantic segmentation pipeline for tracking deforestation using Sentinel-2 satellite imagery. The proposed system integrates automated data acquisition, weakly supervised dataset generation using the Segment Anything Model 3 (SAM3), extensive data augmentation, and supervised training of a DeepLabV3 architecture implemented with TorchGeo. The pipeline enables pixel-level delineation of forested areas and supports the estimation of deforestation boundary evolution and total area loss over time.

Experimental results show that SAM3-generated masks, when refined through morphological post-processing and augmentation, provide an effective proxy for ground truth despite limited manual intervention. The proposed DeepLabV3 model with a SECO-pretrained ResNet50 backbone achieved an Intersection over Union (IoU) of 0.9644 and a Dice coefficient of 0.9816 on the test set. In addition, Local Interpretable Model-agnostic Explanations (LIME) are employed to improve model transparency. This work demonstrates how foundation models and explainable AI can be combined to support scalable and interpretable deforestation monitoring.

Index Terms—Satellite Imagery, Geospatial Data Science, Deforestation Monitoring, Semantic Segmentation, Weakly Supervised Learning, Foundation Models, Explainable AI

A. Introduction

Deforestation is one of the most pressing environmental challenges of the 21st century, contributing to biodiversity loss, ecosystem degradation, and increased greenhouse gas emissions. Accurate and timely monitoring of forest loss is essential for policy-making, conservation strategies, and sustainable land management.

High-revisit, multi-spectral satellite missions such as Sentinel-2 [1] provide frequent and detailed observations of the Earth’s surface, making them particularly suitable for environmental monitoring. Semantic segmentation enables pixel-level classification of forested and non-forested areas, allowing precise tracking of deforestation boundaries and total area loss.

However, supervised deep learning methods require large annotated datasets, which are difficult and expensive to produce in remote sensing contexts. Recent foundation models, such as the Segment Anything Model (SAM), offer new opportunities for weakly supervised annotation by generating high-quality segmentation masks from simple prompts.

The objectives of this project are:

- To design an end-to-end system for deforestation detection using Sentinel-2 imagery
- To generate a high-quality segmentation dataset using SAM3 with minimal manual adjustments
- To train a deep learning model capable of accurately segmenting forested regions
- To provide interpretability for deforestation predictions using explainable AI methods

Contributions:

- A weakly supervised dataset creation pipeline using SAM3
- A TorchGeo-based DeepLabV3 segmentation model with Sentinel-2 pretraining
- A deforestation tracking use case focusing on boundary evolution and area loss
- Integration of LIME for local interpretability of segmentation outputs

B. Related Work

Early deforestation detection methods relied on classical image processing techniques such as thresholding, vegetation indices, and rule-based classification. While computationally efficient, these approaches struggle with complex spatial patterns and varying illumination conditions.

Deep learning approaches, particularly convolutional neural networks such as U-Net, SegNet, and DeepLab, have significantly improved segmentation accuracy by learning hierarchical spatial features. However, their performance depends heavily on large labeled datasets.

To reduce annotation effort, weakly supervised and semi-supervised methods have been explored. Foundation models such as SAM represent a paradigm shift by enabling prompt-based segmentation without task-specific training. Although SAM has shown strong performance on natural images, its application to remote sensing and deforestation monitoring remains relatively underexplored. This project addresses this gap by integrating SAM3 into a complete operational pipeline. Unlike traditional supervised methods that rely on scarce manual labels, this approach leverages the zero-shot generalization capabilities of foundation models. Furthermore, compared to standard weak supervision techniques, the use of SAM3 provides higher fidelity initial masks, reducing the noise typically associated with heuristic label generation.

C. Problem Formulation

The task addressed in this work is **binary semantic segmentation** of forested areas in Sentinel-2 imagery.

- **Input:** Multi-spectral Sentinel-2 images (GeoTIFF format)
- **Output:** Pixel-wise segmentation masks indicating forested regions

Assumptions:

- Sentinel-2 imagery is geometrically and radiometrically corrected
- Textual prompts reasonably describe forested areas

Constraints:

- No manually annotated pixel-level ground truth
- Spatial resolution limited to 10 meters per pixel

Evaluation Metrics:

- Intersection over Union (IoU)
- Dice coefficient
- Qualitative assessment of boundary coherence

D. Dataset

The dataset is constructed from Sentinel-2 imagery obtained from the Copernicus Open Access Hub. The ROIs are selected based on known deforestation hotspots across different continents to ensure diversity in forest types and deforestation patterns (e.g., Amazon rainforest, Australia, Indonesia, tropical Africa). A total of 7 ROIs of variable size were selected. Each of these was then split into multiple smaller tiles of size 224x224 pixels to form the final dataset, composed of a total of 112 patches.

- **Source:** Copernicus Sentinel-2 mission
- **Format:** GeoTIFF

- **Spatial Resolution:** 10 m (RGB bands)

Annotation Process: Key geographic regions exhibiting diverse deforestation patterns (e.g., varying density and boundary complexity) are manually selected. SAM3 is prompted with the text “**green forested areas**” to generate high-precision segmentation masks. Morphological operations are applied to remove noise, fill holes, and smooth boundaries. Afterwards, the masks are visually inspected and corrected as needed.

Dataset Split: (Training: 70% | Validation: 15% | Test: 15%)

Preprocessing:

- Band selection and normalization
- Tiling into fixed-size 224x224 patches (roughly 2.24 km² at 10 m resolution)

Data Augmentation (Applied to training patches to enhance model generalization):

- Horizontal and vertical flips
- Rotations (90°, 180°, 270°)
- Combined geometric transformations (both flips and rotations)

After augmentation, the training set size increased from 78 to 640 patches. Validation and test sets were not augmented (16 patches each).

E. Methodology

The proposed pipeline consists of the following stages:

- 1) **Sentinel-2 Data Acquisition:** Acquisition of multi-spectral imagery from the Copernicus Open Access Hub to serve as the input for the system.
- 2) **Forest Mask Generation:** Utilization of the Segment Anything Model 3 (SAM3) [2] with text prompts (e.g., “green forested areas”) to generate initial weak segmentation masks without manual pixel-level annotation.
- 3) **Mask Refinement:** Application of morphological operations to remove noise and smooth boundaries, followed by manual verification to establish high-quality pseudo-ground truth.
- 4) **Dataset Preparation:** Tiling of satellite images and masks into fixed-size patches, followed by extensive data augmentation to improve model generalization.
- 5) **Supervised Model Training:** Training of a DeepLabV3 architecture with a ResNet50 backbone [3] using the generated weakly supervised dataset.
- 6) **Inference and Analysis:** Application of the trained model to unseen data to estimate deforestation areas,

coupled with LIME-based explainability analysis to interpret model decisions.

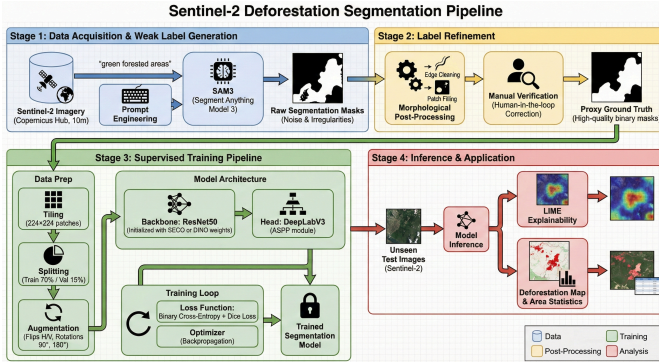


Fig. 1. End-to-end pipeline diagram illustrating data acquisition, mask generation, training, and inference.

Model Architecture: A DeepLabV3 architecture [4], with a ResNet50 backbone [3], implemented with TorchGeo is employed. While architectures like U-Net rely on symmetric encoder-decoder structures with skip connections to preserve fine localization details, DeepLabV3 is often preferred for remote sensing tasks requiring robust contextual understanding. DeepLabV3 utilizes Atrous Spatial Pyramid Pooling (ASPP) to probe the image at multiple scales and fields of view simultaneously via dilated convolutions. This allows the model to effectively capture complex forest patterns and context—distinguishing vegetation from surroundings—which is often more critical in satellite imagery than the pixel-perfect edge sharpness prioritized in biomedical U-Net applications. The ResNet50 backbone acts as the encoder, chosen for its balance of depth and efficiency, and is leveraged here with domain-specific pre-trained weights.

Several experiments are launched, from training from scratch to fine-tuning Sentinel-2 pre-trained backbones. These last ones are leveraged via transfer learning to improve the performance. The weights used are:

- **SECO (Seasonal Contrast)** [5]: pretrained on Sentinel-2 RGB data using seasonal contrastive learning, which helps the model learn to distinguish between our land-cover classes whilst being invariant to seasonal changes.
- **DINO (Self-Distillation with No Labels)** [6]: very powerful variation, pretrained on Sentinel-2 13-bands data using self-supervised learning. Because of that, we had to adapt the loading of the weights to fit our 3-bands input (Red (B04): Index 3, Green (B03): Index 2, Blue (B02): Index 1).

Change Detection and Explainability: To quantify deforestation, the system performs a temporal comparison between two dates (t_1 and t_2) within a defined bounding

box. The DeepLabV3 model generates binary masks for both timestamps. Deforestation is identified by calculating the difference between the masks: pixels classified as forest in t_1 but non-forest in t_2 . The total deforested area is calculated by aggregating these pixels, where each pixel represents $100m^2$ (based on the 10m spatial resolution).

To validate model decisions, Local Interpretable Model-agnostic Explanations (LIME) are generated, producing boundaries that visualize which spectral features most strongly influence the classification.

F. Experimental Framework

- **Frameworks:** PyTorch, TorchGeo, Lightning
- **Key Libraries:** segment-geospatial, torchgeo, leafmap, lightning, alumentations, lime, sentinelhub
- **Hardware:** CUDA-capable GPU (NVIDIA RTX 3050 and NVIDIA RTX 4070)
- **Training Procedure:** Supervised training using SAM3-generated weak labels
- **Loss Function:** Binary Cross-Entropy combined with Dice Loss (to handle class imbalance)
- **Reproducibility:** All scripts and notebooks are provided with fixed parameters

Model Configurations

- 1) **From Scratch:** DeepLabV3 trained from ResNet50 backbone initialized with random weights
- 2) **SECO (Seasonal Contrast)** [5]: ResNet50 backbone pretrained on Sentinel-2 RGB data using seasonal contrastive learning.
 - **100K:** Pretrained on 100,000 Sentinel-2 RGB images.
 - **1M:** Pretrained on 1 million Sentinel-2 RGB images.
- 3) **DINO (Self-Distillation with No Labels)** [6]: ResNet50 backbone pretrained on Sentinel-2 13-bands data using self-supervised learning.

The pretrained weights were used via the following transfer learning techniques:

- **Gradual Unfreezing:** Initially, only the classifier head is trained while the backbone remains frozen. After a few epochs, the entire network is unfrozen and fine-tuned.
- **Discriminative Learning Rates:** Different learning rates are applied to different layers of the network, with lower rates for earlier layers and higher rates for later layers.

As for the run parameters, all the experiments were trained using a batch size of 2 and a maximum of 10 epochs.

Gradual unfreezing was applied after 2 epochs. The learning rate for the backbone was set to $1e-5$, while the classifier head used a learning rate of $1e-3$.

G. Experimental Study

Baseline Models: To establish a performance lower bound, we utilize a DeepLabV3 model trained from scratch (random initialization). Additionally, we define a data baseline using raw SAM3 masks without post-processing to evaluate the quality of the generated ground truth.

Ablation Studies: To validate the design choices, we analyzed the impact of specific pipeline components:

- 1) **Data Preparation:** Comparison between raw SAM3 masks and cleaned masks.
- 2) **Model Initialization:** Evaluation of SECO and DINO pre-training against the random initialization baseline.
- 3) **Training Strategy:** Comparison of Gradual Unfreezing versus Discriminative Learning Rates.

Quantitative Results: When taking a look at the graph displaying the training and validation IoU curves (see Fig. 2), we can observe how the models leveraging Sentinel-2 pre-trained weights outperform the one trained from scratch, but not by a large margin, as the baseline already provides decent results.

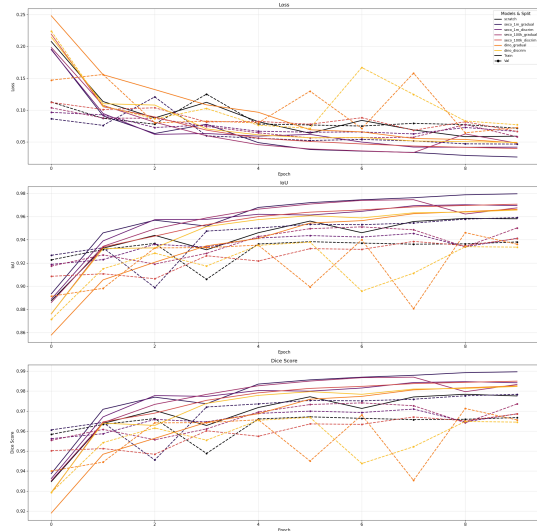


Fig. 2. Training and validation IoU curves for different model configurations.

Another interesting observation is the unstable behavior of the DINO-pretrained models, where the validation IoU fluctuates significantly during training. This could be attributed to the domain gap between DINO’s pre-training data (13-band Sentinel-2) and our 3-band input, leading to less effective feature extraction.

Among all configurations, SECO 1M with Gradual Unfreezing achieves the highest IoU, Dice, and Accuracy scores

on the test set (with the most noticeable performance improvement), demonstrating the importance of season-agnostic pre-training.

TABLE I
QUANTITATIVE COMPARISON OF SEGMENTATION PERFORMANCE ON THE TEST SET.

Model Configuration	IoU	Dice	Accuracy
Scratch	0.9445	0.9710	0.9687
SECO 1M (Gradual Unfreeze)	0.9644	0.9816	0.9800
SECO 1M (Discriminative LR)	0.9558	0.9770	0.9748
SECO 100K (Gradual Unfreeze)	0.9589	0.9787	0.9763
SECO 100K (Discriminative LR)	0.9525	0.9752	0.9733
DINO (Gradual Unfreeze)	0.9446	0.9710	0.9692
DINO (Discriminative LR)	0.9453	0.9715	0.9691

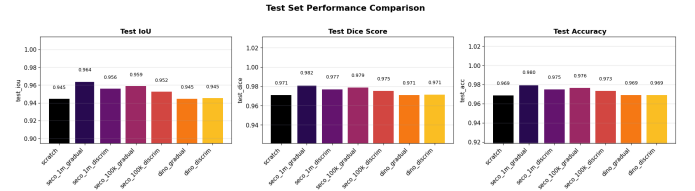


Fig. 3. Bar chart comparing IoU, Dice and Accuracy scores across different model configurations on the test set.

Qualitative Results: Quantitative analysis demonstrates that Sentinel-2 pretrained models (SECO) consistently outperform the baseline trained from scratch in IoU and Dice metrics (see Table I). Visual inspection confirms that the model produces smoother boundaries and reduced noise compared to the raw SAM3 outputs.

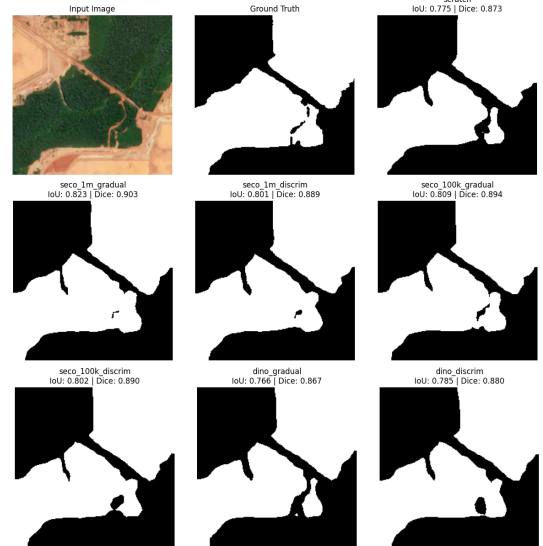


Fig. 4. Qualitative results showing Sentinel-2 input, SAM3 proxy mask, and model prediction.

When examining LIME explanations (see Fig. 5), we observe that the model focuses on relevant boundary regions between forested and non-forested areas. Additionally, because the example displays an input with two rotation augmentations next to each other, we can see how the model is changing its focus accordingly, which indicates that the model is not overfitting to specific spatial patterns, and highlights the effectiveness of the augmentation strategy.

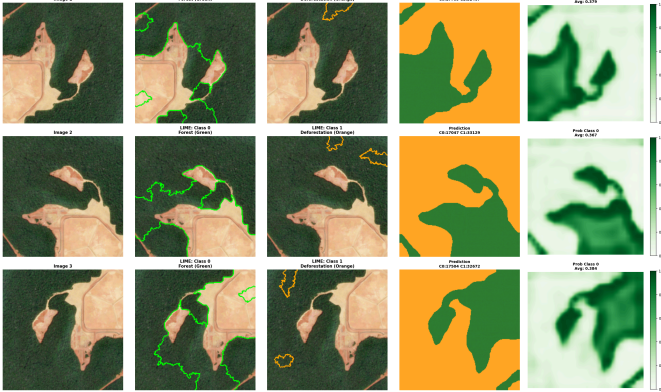


Fig. 5. LIME explanations highlighting important regions influencing the model's forest classification.

Deforestation Detection Over Time: Although our current method for deforestation detection works reasonably well, there is still room for improvement. As shown in Fig. 6, the model successfully identifies the deforestation start in the first example (top), but struggles with the second one (bottom), where it's flagged earlier than it should have been.

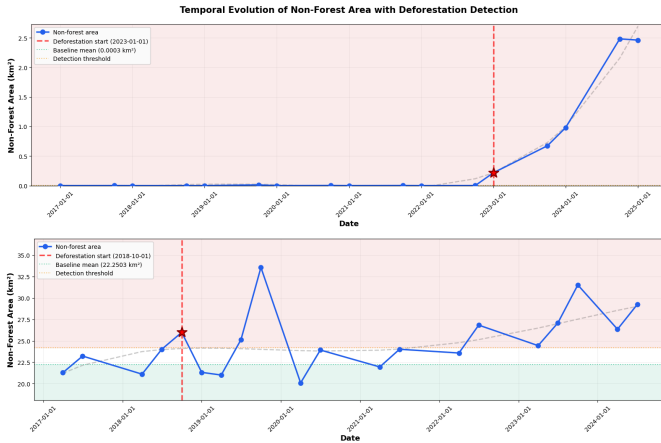


Fig. 6. Deforestation detection results on two different patches over time.

H. Conclusions and Future Work

Summary of Findings: The Segment Anything Model 3 (SAM3) has proven to be a versatile tool for generating high-quality segmentation masks in the context of deforestation monitoring using Sentinel-2 imagery. By leveraging text prompts, SAM3 enables the creation of weakly supervised

datasets with minimal manual intervention, significantly reducing annotation costs.



Fig. 7. Examples of SAM3-generated masks before and after morphological post-processing. left: input image; center: raw SAM3 mask; right: cleaned SAM3 mask.

The DeepLabV3 model, particularly when initialized with SECO-pretrained weights, demonstrates strong performance in segmenting forested areas. The use of Sentinel-2 specific pre-training significantly enhances the model's ability to generalize from limited weakly supervised data. The Gradual Unfreezing strategy further boosts performance by allowing the model to adaptively learn relevant features.

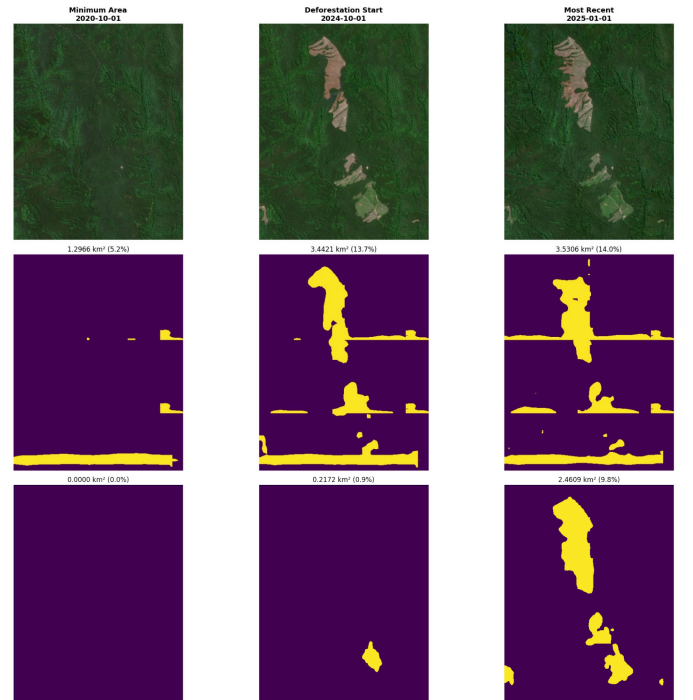


Fig. 8. Examples of results when training directly on unprocessed SAM3 masks. Same patch of Sentinel-2 imagery in different dates. top: input images; middle: predictions trained on raw SAM3 masks; bottom: predictions trained on cleaned SAM3 masks.

Limitations: Despite the promising results, several limitations remain. First, the reliance on weak supervision means the model inherits biases and artifacts from SAM3, particularly in transition zones where the “green forested area” prompt may be ambiguous. Second, the lack of pixel-perfect ground truth makes it difficult to distinguish between model errors and annotation errors during evaluation. Finally, the

current approach is binary; distinguishing between different forest types or degradation levels remains a challenge.

Future Work:

- Generalization of land-cover segmentation beyond binary forest/non-forest classes
- Fine-tuning SAM on remote sensing data
- Integration of multi-temporal Sentinel-2 imagery
- Iterate on explainability methods for deeper insights
- Iterate on deforestation change-detection-over-time method improvements
- Integration of CLAY (geospatial embeddings) to automatically detect deforested areas
- Custom segmentation for temporal tracking of deforestation

I. References

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